Atharva Mohite

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EDUCATION

Georgia Institute of Technology *Aug 2026 – May 2028 (Expected)*
Master of Science in Computer Science, Machine Learning Atlanta, GA
Relevant Coursework: Machine Learning, Big Data Systems, Data Visualization and Analytics

Sardar Patel Institute of Technology *Aug 2019 – May 2023*
Bachelor of Technology (BTech) in Computer Science | **CGPA**: 9.39/10 Mumbai, India

EXPERIENCE

JPMorgan Chase *Jan 2026 – Aug 2026*
Software Engineer II

- Owned cross-service design and production readiness for a new business line across **7 microservices** on a multi-tenant settlement platform, scaling throughput to **5M+** trades per day.
- Built an **agentic SDLC** pipeline that autonomously translated Jira requirements into code changes across a multi-service architecture, reproduced issues with local infrastructure, validated via automated tests, and raised developer-ready PRs for review, with **75%** PRs merged with minimal edits.
- Built a regression framework using **LLM agents** that generates and diagnoses multi-service test workflows, automating **300+** scenarios and cutting manual test effort by **90%**; adopted by 30+ engineers across 5 teams.
- Designed an aggregation engine in Spring Boot reducing high-volume trade streams into consolidated settlement instructions across configurable dimensions, cutting down instruction volume by **70%**.

JPMorgan Chase *Feb 2023 – Dec 2025*
Software Engineer I

- Led extraction of a high-throughput microservice from a legacy trade settlement monolith, processing **1M+** messages per day, with an event-driven, non-blocking stack with Kafka and Spring WebFlux.
- Designed distributed end-of-day orchestration coordinating 3 services to close and settle the day's book within a **30**-minute cutoff, handling race conditions, ordering dependencies, and concurrent state transitions with idempotent replay on failure.
- Architected a materialized datastore using **Elasticsearch**, aggregating and rehydrating data from multiple source systems to provide **sub-150ms** complex search over **750k** records, enabling the decommission of **4** legacy systems.
- Migrated 15 legacy applications to an in-house **Kubernetes** platform, saving **$450k** annually in infrastructure costs and reducing deployment time by **50%**.

JPMorgan Chase *Jan 2022 – Jul 2022*
Software Engineer Intern

- Developed a cloud resource discovery framework that helped discover **10k+** resources, consolidating AWS Config, Splunk, Terraform, and tagging APIs into a unified inventory using Qlik Sense dashboards for tracking resource ownership, enabling data-driven cleanup decisions and improving infrastructure visibility for the DevOps team.

SKILLS

Languages: Python, Java, TypeScript, SQL
AI/ML: LLM agents, RAG, LangChain, LangGraph, MCP, PyTorch, Hugging Face Transformers, vLLM, evals and prompt design
Backend & Infra: Spring Boot, Flask, Node.js, Django, Relational Databases (Oracle), Elasticsearch, AWS, Terraform, Kubernetes, Docker, Jenkins, Shell Scripting, React, Next.js
Other: System design (scalable and distributed systems), Event-driven architecture (Kafka), RESTful and GraphQL API design, Cloud Architecture, Testing (unit, integration, E2E), CI/CD pipelines, Design Patterns, Monitoring and Observability (Splunk, Grafana).

PROJECTS

Splice | Code | **PyTorch, Transformers, vLLM**

- Cut time to first token **1.7×** and raised throughput **52%** at 90% memory reuse, and up to **2.3×** with all memories precached, by re-rotating stored KV through vLLM's KV connector API without engine changes.
- Showed uncorrected reuse halves answer quality and the best of 3 published fixes needs **~70%** recompute to recover it, by benchmarking them with Mistral-7B on **1,986** LoCoMo questions.

Canary | Code | **PyTorch, LightGBM, vLLM**

- Cut agent compute **46%** at a **2%** loss of successful runs, nearly double the best fixed turn limit, by stopping likely failures early.
- Showed most early failure signal is task difficulty, AUROC **0.70** across all runs vs. **0.54** between attempts at the same issue, by training LightGBM and GRU monitors on **147k** SWE-agent and OpenHands trajectories.

RESEARCH

Leveraging LLMs for Video Querying | Paper Link | IEEE CERA 2023

- GPT-4 matched **56%** of event timestamps exactly and **85%** within 2 min by querying BMT and Vid2Seq captions of surveillance video.

SecurePark - a vehicle intrusion detection system | Paper Link | IEEE ASIANCON 2022

- Trained a YOLOv5s license-plate detector and wired it to OCR, authorization checks, alerts, and a dashboard.